

SEED ROUND

SYNTHEGY

Strategy-Aware AI for Synthetic Chemistry

Seeking: \$3-5M seed | Valuation: \$15-25M | Use of funds: 3 paid pilots +
proprietary data flywheel

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The Problem

Pharma companies spend **\$2.6 billion per approved drug**. 90% of candidates fail in clinical trials. Much of that failure is predictable at the chemistry stage.

When a med-chem team picks a synthesis route to advance, they make a **\$5-10M decision** with spreadsheets, gut feel, and a 2-hour meeting.

The data needed to decide empirically exists across 12 different public databases. No one connects them.

The Solution

SyntheGy turns a 3-day multi-database lookup into a **2-minute automated report**.

The chemist types a natural-language instruction. The platform fetches PubChem properties, ChEMBL bioactivity, ORD experimental reactions, RDKit ADMET descriptors, Europe PMC literature confidence, RCSB PDB structures, KEGG pathways, OpenTargets disease associations, Google Patents, openFDA adverse events, and WHO health indicators -- then feeds it all to an LLM Strategic Evaluator that scores the route and explains its reasoning.

"Bench to bedside, in one platform."

The Knowledge Graph

The only platform that traverses **4 databases in 1 API call**:

Compound (PubChem) -> Targets (ChEMBL + gene extraction)
-> Pathways (KEGG) -> Diseases (OpenTargets, with scores)

Aspirin -> Cyclooxygenase (PTGS2) -> 15 diseases (rheumatoid arthritis 0.635, gout 0.631, migraine 0.631, osteoarthritis 0.623, myocardial infarction...) in **under 2 seconds**.

Provisional patent filed on the traversal method. This is the moat.

Architecture & Traction

7	12	300	30+
tiers	data sources	patients	API endpoints

4 Bun/Hono microservices (ports 3001-3005) + Next.js 16 frontend. All 12 data sources are **free, public, no API keys**. Marginal cost per query: effectively zero. All code on GitHub (11 commits, full history).

The Data Flywheel

Every accept/revise signal trains the next recommendation. After 1,000 real outcomes, the evaluator is better than any rule-based system because it is trained on **your customers' actual lab results**.

More customers -> more outcomes -> better predictions -> more customers.

The active-learning feedback loop already extracts keyword patterns from accepted vs revised instructions. After 100+ signals per org, the evaluator becomes organisation-specific -- and that is sticky. Switching cost = the moat.

Clinical Layer: Bench to Bedside

300 synthetic patients across 6 diseases (RA, migraine, gout, OA, UC, asthma) with 50-year epidemiological basis (1975-2025). Real government data integration (openFDA + WHO). Outcome prediction model. CDISC SDTM regulatory export. RWE CSV import pipeline.

"One function changes; the platform adapts." -- replace the synthetic generator with a CSV import when real patient data arrives. The entire analysis pipeline stays identical.

Market Opportunity

Segment	Size	Synthegy's Play
Chemistry software TAM	\$2B	SaaS at \$50-150K/year per team
AI in drug discovery	\$8B by 2030	Own the route-selection decision moment
Late-stage drug failures	\$50B/year	Predict failures at the chemistry stage
Comparable exits	\$1.5-2B	Exscientia (\$1.5B), Recursion (\$2B), Insitro (\$2B)

Use of Funds (\$3-5M)

40% -- 3 paid pilots (\$15K each, convert 2 to \$50K/year annuals). Target: mid-size pharma (50-500 R&D; scientists). Goal: \$500K ARR + 3 case studies by month 9.

30% -- Proprietary data flywheel. Build the route-outcome capture pipeline. Every time a chemist runs the evaluator and synthesises the route, record whether it worked. After 1,000 real outcomes, the evaluator is better than any rule-based system.

20% -- Hospital/CRO partnership. Partner with 1 hospital for de-identified RA or asthma data. Swap the synthetic generator for a CSV import. First real-world evidence platform deployment.

10% -- IP + legal. File the full (non-provisional) patent. Incorporate. Trademark 'Synthegey'.

The Ask

Raising \$3-5M seed at a \$15-25M valuation.

12-month milestones: \$500K ARR, 3 paying pilots, 1 hospital data partnership, provisional patent filed, first Compound Advancement Report delivered to a pharma partner.

Investors: a16z Bio + ASC, Data Collective, or chemistry-specialist funds.

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Live platform:

preview-chat-6cbbdd64-b04b-40df-a2f3-12616b64fcbf.space-z.ai

GitHub: github.com/testdemoqwenai2025-creator/syntheyg